

* Total prime numbers between 1-50 $\rightarrow$ 15

50-100 $\rightarrow$ 10

100-150 $\rightarrow$ 10

150-200 $\rightarrow$ 11

200-250 $\rightarrow$ 7

250-300 $\rightarrow$ 10

Co-prime numbers :

- * Two numbers are co-prime if their HCF is 1.
- * Two prime numbers are always co-prime to each other.
- * Any two consecutive integers are always co-prime.
- * Face value of 3 in 56738 is 3.
- * Place value of 3 in 56738 is 30.

Operation	Rational, Rational	Rational, Irrational	Irrational, Irrational
Addition	Always rational $2 + 3 = 5$	Always irrational $2 + \sqrt{3}$	Can be rational or irrational $\sqrt{2} + (-\sqrt{2}) = 0$ $\sqrt{2} + \sqrt{2} = 2\sqrt{2}$
Subtraction	Always rational $3 - 2 = 1$	Always irrational $3 - \sqrt{2}$	Can be rational or irrational $\sqrt{3} - \sqrt{3} = 0$ $\sqrt{3} - \sqrt{2}$
Multiplication	Always rational $2 \times 3 = 6$	Always irrational Except if rational number is 0 $2 \times \sqrt{3} = 2\sqrt{3}$ $0 \times \sqrt{3} = 0$	Can be rational or irrational $2\sqrt{2} \times \sqrt{2} = 4$ $\sqrt{2} \times \sqrt{3} = \sqrt{6}$
Division	Always rational (if divisor $\neq 0$) $\frac{3}{4} = 0.75$	Always irrational Except if dividend is 0. $\frac{2}{\sqrt{3}}$ $\frac{0}{\sqrt{3}} = 0$	Can be rational or irrational $\frac{2\sqrt{2}}{\sqrt{2}} = 2$ $\frac{\sqrt{2}}{\sqrt{3}} = \sqrt{\frac{2}{3}}$

Successive Division

* A number 219 is successively divided by 6, 2, 4, 3

$$\begin{array}{r} 6 \overline{) 219} \quad (36 \\ \underline{18} \\ 39 \\ \underline{36} \\ 3 \end{array}$$

$$\begin{array}{r} 2 \overline{) 36} \quad (18 \\ \underline{2} \\ 16 \\ \underline{16} \\ 0 \end{array}$$

$$\begin{array}{r} 4 \overline{) 18} \quad (4 \\ \underline{16} \\ 2 \end{array}$$

$$\begin{array}{r} 3 \overline{) 4} \quad (1 \\ \underline{3} \\ 1 \end{array}$$

$$\begin{array}{r} 6 \xrightarrow{+} 3 \\ \quad \times \\ 2 \xrightarrow{+} 0 \\ \quad \times \\ 4 \xrightarrow{+} 2 \\ \quad \times \\ 3 \xrightarrow{+} 1 \\ \quad \times_1 \\ \quad \nwarrow \text{last quotient} \end{array}$$

$$3 \times 1 + 1 = 4$$

$$4 \times 4 + 2 = 18$$

$$18 \times 2 + 0 = 36$$

$$36 \times 6 + 3 = 219$$

* If they don't give the last quotient, we can assume it as 0

Divisibility Rule of 2, 4, 8, 16

- * $2 (2^1)$: Last digit is divisible by 2 or is 0.
- * $4 (2^2)$: Last 2 digits are divisible by 4 or are 00.
- * $8 (2^3)$: Last 3 digits are divisible by 8 or are 000.
- * $16 (2^4)$: Last 4 digits are divisible by 16 or are 0000.

Divisibility Rule of 3, 9

- * 3 : Sum of digits is divisible by 3.
- * 9 : Sum of digits is divisible by 9.

Divisibility Rule of 5, 25, 125

- * 5 (5^1) : Last digit is 0 or 5.
- * 25 (5^2) : Last 2 digits are 00 or divisible by 25.
- * 125 (5^3) : Last 3 digits are 000 or divisible by 125.

Divisibility Rule of Numbers having known

- * 6 : Number is divisible by both 2 and 3.
- * 24 : Number is divisible by both 8 and 3.
- * 72 : Number is divisible by both 8 and 9.
- * 88 : Number is divisible by both 8 and 11.

Note:- We have to take only co-primes.

Divisibility Rule of 7, 11, 13

Check 88440077061

$$\begin{array}{c} \text{88} \mid \text{440} \mid \text{077} \mid \text{061} \end{array}$$

$$\begin{array}{r} 061 \\ 440 \\ \hline (+) \\ 501 \end{array}$$

$$\begin{array}{r} 077 \\ 88 \\ \hline (+) \\ 165 \end{array}$$

$$\begin{array}{r} 501 \\ 165 \\ \hline (-) \\ 336 \end{array}$$

Now we need to check whether 336 is divisible by 7, 11 or 13.

Divisibility Rule of 11

Check 5453987

5|4|5|3|9|8|7

$$7 + 9 + 5 + 5 = 26$$

$$8 + 3 + 4 = 15$$

$$\begin{array}{r} \leftarrow \\ \hline 11 \end{array}$$

The result is divisible $\swarrow$ 11 or not.

Note :- Number in the form of abcabc are always divisible by 7, 11, 13 and 1001.

Remainder Theorem

$$\frac{17}{3} \rightarrow 2$$

$$\frac{17^2}{3} = \frac{17 \times 17}{3} \Rightarrow \frac{2^2}{3} \rightarrow \frac{4}{3} \rightarrow 1$$

$$\frac{35}{37} \rightarrow -2 \rightarrow 35$$

$$\frac{35^2}{37} = \frac{35 \times 35}{37} \Rightarrow \frac{(-2)^2}{37} \rightarrow \frac{4}{37} \rightarrow 4$$

$$\frac{35^3}{37} = \frac{35 \times 35 \times 35}{37} \Rightarrow \frac{(-2)^3}{37} \rightarrow \frac{-8}{37} \rightarrow 29$$

$$\frac{35 \times 38 \times 39}{37} \rightarrow \frac{-4}{37} \rightarrow 33$$

$$\frac{35 + 38 + 39}{37} \rightarrow \frac{1}{37} \rightarrow 1$$

* $\frac{a^{p-1}}{p} \rightarrow \text{Remainder} = 1$ where p is prime number,

a and p are co-primes.

Ex: $\frac{8^{36}}{37} \rightarrow 1$

$$\frac{8^{13}}{37} = \frac{8^{12} \cdot 8}{37} = \frac{8^{36} \cdot 8^{36} \cdot 8}{37} \rightarrow \frac{8}{37} \rightarrow 8$$

$$\frac{5^{50}}{17} = \frac{5^{48} \cdot 5^2}{17} = \frac{5^{16} \cdot 5^{16} \cdot 5^{16} \cdot 5^2}{17} \rightarrow \frac{5^2}{17} \rightarrow \frac{25}{17} \rightarrow 8$$

Number of zeroes

* Let $N = 2^p \times x^a \times y^b \times z^c \times 5^q$

Number of zeroes in $N = \min(p, q)$

* To find the number of zeroes in $n!$, we divide n by 5 repeatedly until the result is less than 5 and then add all the quotients. $\left(\left\lfloor \frac{n}{5} \right\rfloor + \left\lfloor \frac{n}{25} \right\rfloor + \left\lfloor \frac{n}{125} \right\rfloor + \dots\right)$

Ex: $\frac{50!}{5} \rightarrow 10 + 2 \rightarrow 12 \text{ zeroes}$

Factors

$$* a^m - b^m \text{ (m is even)} \rightarrow (a+b), (a-b)$$

$$* a^m + b^m \text{ (m is odd)} \rightarrow a+b$$

$$* a^m - b^m \text{ (m is odd)} \rightarrow a-b$$

$$* a^m + b^m \text{ (m is even)}$$

$$a^{12} + b^{12} \rightarrow (a^4)^3 + (b^4)^3 \rightarrow a^4 + b^4$$

$$(a^m + b^m \text{ (m is odd)})$$

Number of factors

$$120 = 2^3 \times 3^1 \times 5^1$$

* Total number of factors (including 1 and itself)

$$= (3+1)(1+1)(1+1) = 4 \times 2 \times 2 = 16$$

* Total number of even factors = $(3)(1+1)(1+1)$

$$= 3 \times 2 \times 2 = 12$$

* Total number of odd factors = $(1)(1+1)(1+1) = 1(2)(2) = 4$

* Total number of prime factors = $3 + 1 + 1 = 5$

* Sum of all factors = $(2^0 + 2^1 + 2^2 + 2^3) \cdot (3^0 + 3^1) \cdot (5^0 + 5^1)$

* Sum of even factors (ignore 2^0)

$$= (2^1 + 2^2 + 2^3) \cdot (3^0 + 3^1) \cdot (5^0 + 5^1)$$

* Sum of odd factors = $(3^0 + 3^1) \cdot (5^0 + 5^1)$

Units Digit of powers

0 → Always 0

1 → Always 1

2 → 2, 4, 8, 6, ...

3 → 3, 9, 7, 1, ...

4 → 4, 6, 4, 6, ...

5 → Always 5

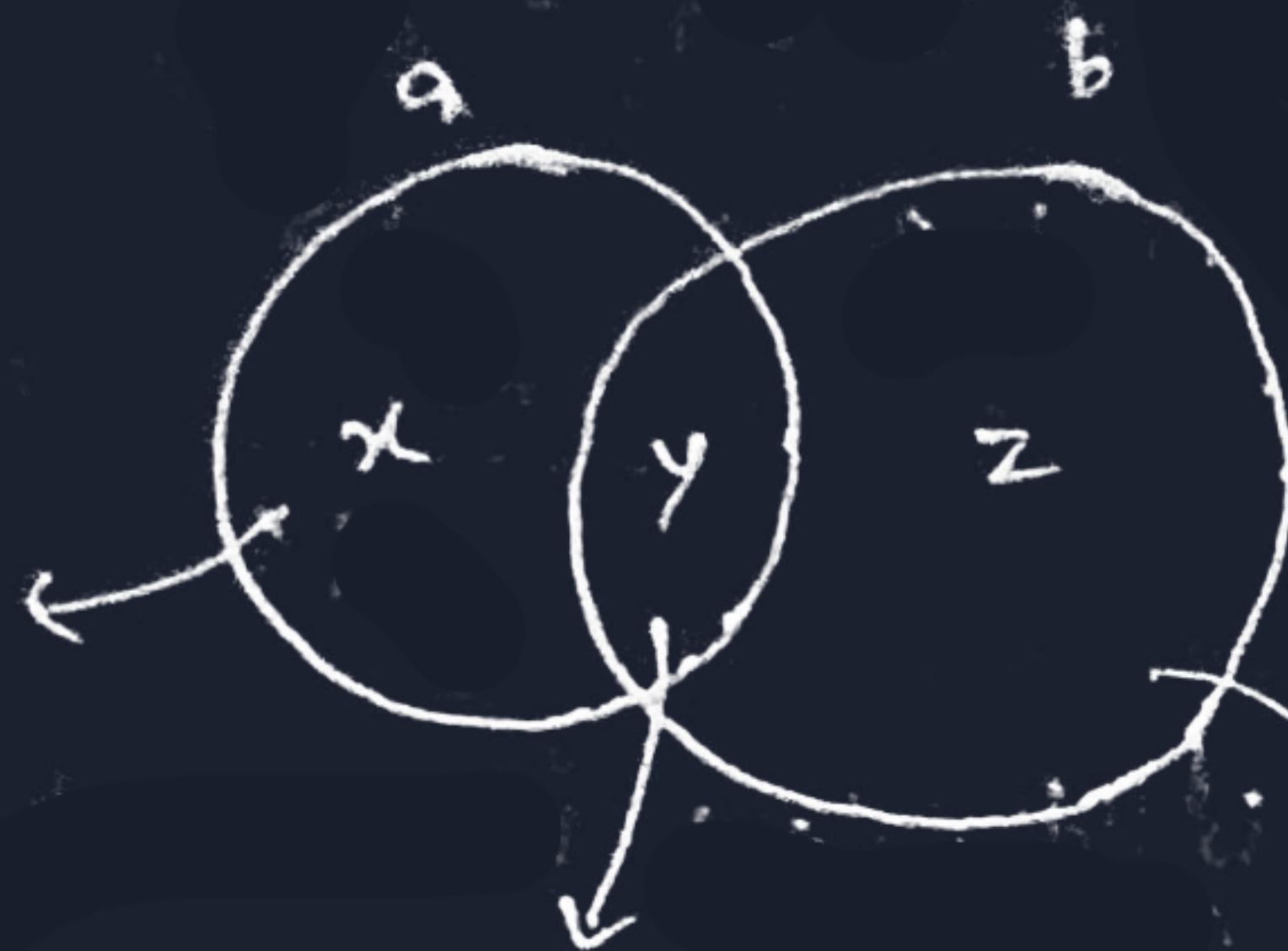
6 → Always 6

7 → 7, 9, 3, 1, ...

8 → 8, 4, 2, 6, ...

9 → 9, 1, 9, 1, ...

Divisibility by a or b, a and b



divisible by a
but not by b

divisible by a & b

divisible by b but not by a

$x+y \rightarrow$ divisible by a

$y+z \rightarrow$ divisible by b

$x+y+z \rightarrow$ divisible by a or b

$(x+y) + (y+z) - (2y) = x+z \rightarrow$ divisible by a or b but not
by a & b.

Recurring Decimal

$$0.\overline{23} = \frac{23}{99}$$

$$0.5\overline{29} = \frac{529 - 5}{990} = \frac{524}{990}$$

$$0.83\overline{51} = \frac{8351 - 8}{9990} = \frac{8343}{9990}$$

$$2.53\overline{2} = 2 + 0.53\overline{2} = 2 + \frac{532 - 53}{900} = 2 + \frac{479}{990}$$

Identification of Perfect Square

- * Last digit is 1, 4, 5, 6, 9 or 0, then the number may or may not be a perfect square.
- * Last digit is 2, 3, 7, 8, then the number is not a perfect square.
- * Last two digits of the number should match with the last two digits of the numbers 1^2 to 24^2 .
- * Digital sum should be 1, 4, 9 or 7 (1947).